"NutriScan: A Database-Driven Food Nutrition Analysis and Healthier Alternative Recommendation System"



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(Computer Science and Engineering)

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RECOMMENDATION

We are pleased to recommend that the project report of phase-I entitled **"NutriScan: A Database-Driven Food Nutrition Analysis and Healthier Alternative Recommendation System"** submitted by 0801CS25D03 - Divy Arora, 0801CS25D07 - Pawan Mewada, 0801CS25D09 - Samiya Akhtar, 0801CS25D10 - Sudhanshu Suryawanshi and 0801CS241127 - Santosh, students of B.Tech. III year may be accepted in partial fulfillment of the degree of **Bachelor of Technology, Computer Science & Engineering** of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the session **2026-27**.

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CERTIFICATE

This is to certify that the project report entitled **"NutriScan: A Database-Driven Food Nutrition Analysis and Healthier Alternative Recommendation System"** submitted by 0801CS25D03 - Divy Arora, 0801CS25D07 - Pawan Mewada, 0801CS25D09 - Samiya Akhtar, 0801CS25D10 - Sudhanshu Suryawanshi and 0801CS241127 - Santosh, students of III year B.Tech. **(Computer Science & Engineering) in the year 2026-27 of the Institute,** is a satisfactory account of their Project Phase-I work based on the syllabus.

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DECLARATION

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We further declare that, to the best of our knowledge, this work does not contain any part of any work which has been submitted for the award of any degree or any other work either in this University or in any other University/website without proper citation.

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ABSTRACT

Packaged foods such as biscuits, chips, chocolates, cereals and instant noodles are now part of everyday diets, yet most consumers find it difficult to judge their nutritional quality from the label. Nutrition tables are dense, expressed in different units, and rarely compared across brands at the point of purchase. This project presents NutriScan, a mobile application that converts a product barcode into a simple, transparent and actionable nutritional assessment.

NutriScan follows a database-driven architecture. Product records from the Open Food Facts dataset are cleaned, standardized and stored in a PostgreSQL database. When a user scans a barcode, a Node.js and Express.js backend retrieves the matching record, and a rule-based scoring engine rates the product from 0.5 to 5 stars using the Indian Nutrition Rating (INR) methodology proposed by FSSAI in 2022. The rating is accompanied by a nutritional breakdown, ingredient information, positive attributes, potential concerns and a plain-language explanation of how the rating was obtained. No machine-learning model is used, so every rating can be traced to explicit rules.

The central feature of the system is the recommendation of healthier alternatives: products of the same category with a higher star rating and sufficiently reliable data are ranked and presented, together with the reasons they rate better. Users can also compare products side by side, search by name, maintain scan history and favorites, and set preferences such as lower sugar or higher protein. If a product is absent from the database or its data are incomplete, the application says so rather than inventing values.

The report covers the problem analysis, requirements, data preparation, system architecture, database design, scoring methodology, recommendation algorithm, comparison module and API design of the proposed system. NutriScan does not replace professional dietary advice; it aims to make nutritional information more accessible during everyday shopping.

Keywords: Nutrition analysis, barcode scanning, Indian Nutrition Rating, healthier alternatives, Open Food Facts, PostgreSQL, React Native, rule-based recommendation.

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LIST OF SYMBOLS

Symbol	Description
S	Overall Health Score of the initial design, 0 to 100 (Section 4.6); in Section 4.7, S denotes saturated-fat points
i	Index of a scoring parameter (sugars, saturated fat, sodium, energy, fiber, protein)
$\mathbf{x_i}$	Measured value of parameter i per 100 g (or 100 ml) of product
$\mathbf{lo_i}$	Value of parameter i at or below which the best sub-score is awarded (for nutrients to limit)
$\mathbf{hi_i}$	Value of parameter i at or above which the worst sub-score is awarded (for nutrients to limit)
$\mathbf{t_i}$	Normalized position of x_i between the thresholds, $t_i = \min(1, \max(0, (x_i - lo_i) / (hi_i - lo_i)))$
$\mathbf{s_i}$	Sub-score of parameter i, in the range 0 to 1
$\mathbf{w_i}$	Weight of parameter i; the weights sum to 1
Σ	Summation operator

LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CSV	Comma-Separated Values
DB	Database
EAN	European Article Number
ETL	Extract, Transform, Load
GTIN	Global Trade Item Number
HTTP(S)	HyperText Transfer Protocol (Secure)
JSON	JavaScript Object Notation
JWT	JSON Web Token
kcal	Kilocalorie
MVP	Minimum Viable Product
NCD	Non-Communicable Disease
OFF	Open Food Facts
ODbL	Open Database License
ORM	Object-Relational Mapping
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
UPC	Universal Product Code
UX	User Experience
WHO	World Health Organization
FSSAI	Food Safety and Standards Authority of India
ICMR-NIN	Indian Council of Medical Research - National Institute of Nutrition
FOPNL	Front-of-Pack Nutrition Labelling
SDK	Software Development Kit
FV	Fruit and Vegetables
INR	Indian Nutrition Rating
NLM	Nuts, Legumes and Millets

Chapter 1

INTRODUCTION

This chapter introduces the NutriScan project. Section 1.1 gives the preamble, Section 1.2 explains the need for the project, Section 1.3 states the problem, Section 1.4 lists the objectives, Section 1.5 outlines the proposed approach, and Section 1.6 describes how the rest of the report is organized.

1.1 Preamble

Packaged and processed foods have become a routine part of daily consumption. Biscuits, chips, chocolates, breakfast cereals, beverages, instant noodles and protein bars are bought quickly, often influenced by taste, price, advertising and brand familiarity rather than by nutritional composition. The nutritional information is printed on every pack, but reading it requires time, a basic understanding of nutrients, and the ability to compare values across products that may be expressed per serving, per 100 g or per pack.

NutriScan is a mobile application designed to remove this friction. The user scans the barcode of a packaged food product, and the application retrieves the product from a curated food database, calculates a nutritional star rating from 0.5 to 5 stars based on the proposed Indian Nutrition Rating (INR), and presents the nutritional breakdown, ingredient list, positive attributes and potential concerns in a simple dashboard. Beyond scoring, NutriScan recommends better-rated alternatives from the same product category and allows products to be compared side by side.

The system is deliberately database-driven and rule-based. Every supported product has an entry in the database, and every score is produced by an explicit scoring methodology rather than a machine-learning prediction. If a product is missing or its data are incomplete, the application reports this honestly instead of fabricating information.

1.2 Need of the Project

Diet-related non-communicable diseases such as obesity, type 2 diabetes and hypertension are widely reported as growing public-health concerns in India and elsewhere [10]. Excess intake of sugars, saturated fat and sodium is associated with these conditions, and international guidance recommends limiting each of them [8], [9], [17]. Packaged foods can be significant contributors to such intake, and consumers have limited tools to assess them quickly while shopping.

Existing front-of-pack schemes such as Nutri-Score [1] and the Health Star Rating [3] show that a simple summary indicator helps consumers, but they are labels printed by manufacturers or regulators and do not, by themselves, tell a shopper which alternative product is better. Several mobile applications scan barcodes, yet many do not expose how the rating is derived, are not tuned to products sold in the local market, or do not combine scoring with category-wise alternatives and comparison in one workflow. NutriScan is needed to fill this gap with a transparent, explainable and practical shopping companion.

1.3 Problem Statement

To design and develop a mobile application that allows a user to scan the barcode of a packaged food product and obtain, from a verified database, (i) a transparent star rating based on the proposed Indian Nutrition Rating, with a breakdown of how it was obtained, (ii) a nutritional and ingredient analysis highlighting positives and concerns, (iii) a ranked list of healthier alternatives within the same product category, and (iv) a side-by-side comparison of products, while explicitly handling missing products and incomplete nutritional data without fabricating values.

1.4 Project Objectives

The objectives of the project are as follows:

1. To build a React Native (Expo) mobile application that scans product barcodes and displays results in an easy-to-understand dashboard.
2. To acquire, clean, standardize and store packaged-food data from Open Food Facts in a PostgreSQL database.
3. To design and implement a rule-based scoring engine based on the Indian Nutrition Rating methodology proposed by FSSAI, returning a star rating and a score breakdown.
4. To recommend better-rated alternatives from the same category using only products with sufficiently reliable data.
5. To provide product comparison, product-name search, category-based discovery, scan history, favorites and preference-based filtering.
6. To handle missing products and incomplete data transparently and to validate the scoring methodology against an independent reference.

1.5 Proposed Approach

The proposed solution has four cooperating parts. First, a data pipeline downloads the Open Food Facts dataset, removes duplicates, standardizes units, handles missing values and loads the cleaned records into PostgreSQL. Second, a Node.js and Express.js backend exposes REST endpoints for barcode lookup, search, scoring, alternatives and comparison. Third, a scoring engine inside the backend applies the INR methodology: it awards baseline points for energy, saturated fat, total sugars and sodium, subtracts eligible positive points for fruit and vegetables, nuts, legumes and millets, fibre and protein, and returns a star rating with a score breakdown. Fourth, the React Native application captures barcodes through the phone camera and presents the results.

Scope. The system covers packaged foods available in the database. It provides nutritional information and comparison for everyday purchasing decisions. **Limitations.** NutriScan does not diagnose medical conditions, does not replace a dietitian, and does not claim that a product is medically suitable for any individual. Its accuracy depends on the quality and currency of the underlying data.

1.6 Organization of the Report

Chapter 2 reviews the literature on nutrient profiling, front-of-pack labelling, existing applications and open food data. Chapter 3 describes the dataset and its preparation. Chapter 4 presents the requirement analysis, system architecture, database design, scoring methodology, recommendation algorithm, comparison module and API design. The appendix contains the database schema, followed by the references.

Chapter 2

LITERATURE REVIEW

This chapter reviews the nutrient-profiling models and labelling systems that inform NutriScan's scoring design, the dietary guidance behind its parameters, existing consumer applications, and the Open Food Facts database used as the data source.

2.1 Nutrient Profiling and Front-of-Pack Labelling

Nutrient profiling is the science of classifying or ranking foods according to their nutritional composition. Profiling models underpin many labelling schemes and regulatory measures.

2.1.1 Nutri-Score

Nutri-Score is a five-colour front-of-pack label (A to E) adopted in France and several other European countries [1]. It computes points per 100 g of product: unfavorable components such as energy, sugars, saturated fat and sodium add negative points, while favorable components such as fiber, protein and the proportion of fruit, vegetables and legumes reduce them. The final points are mapped to a letter grade. Its main lessons for NutriScan are the use of a per-100 g basis, the combination of nutrients to limit and nutrients to encourage, and the value of a simple summary indicator.

2.1.2 UK Nutrient Profiling Model

The UK Ofcom nutrient profiling model, developed to classify foods and drinks for advertising to children, also scores energy, saturated fat, sugars and sodium against fiber, protein and fruit, vegetable and nut content [2]. Like Nutri-Score, it demonstrates that a transparent additive points system can discriminate between foods using a small number of nutrients.

2.1.3 Health Star Rating

The Health Star Rating is a front-of-pack system used in Australia and New Zealand that awards between half a star and five stars [3]. It is category-aware, which supports the design decision in NutriScan to avoid comparing, for example, beverages with biscuits using identical thresholds.

2.1.4 Nutrition Labelling in India

In India, the Food Safety and Standards (Labelling and Display) Regulations, 2020 prescribe the information, including nutritional information, that must appear on packaged food labels [4]. Interpretive front-of-pack labelling has been under discussion, and the current regulatory position should be verified against the latest

FSSAI notifications. Because the printed table is the common denominator across brands, it forms the natural input for a scoring application. The FSSAI draft notification of 2022 proposed an Indian Nutrition Rating (INR) that expresses the profile of eligible packaged foods on a scale of 0.5 to 5 stars [18]; NutriScan adopts this methodology as the basis of its current score calculation (Section 4.7).

2.2 Dietary Guidance Underpinning the Scoring Parameters

The parameters used by NutriScan correspond to nutrients for which health authorities provide guidance:

- **Sugars:** the WHO recommends that free sugars be limited to less than 10% of total energy intake, with a further reduction to below 5% suggested for additional benefit [8].
- **Sodium:** the WHO recommends reducing sodium intake to less than 2 g per day, equivalent to about 5 g of salt [9].
- **Saturated fat:** the WHO recommends limiting saturated fat to less than 10% of total energy intake [17].
- **Fiber and protein:** national and international dietary guidelines, including the Indian guidelines, treat adequate fiber and protein as desirable components of the diet [10].
- **Energy density:** high energy density per 100 g is commonly used in profiling models as an unfavorable indicator [1], [2].

These guidelines apply to total daily diets rather than to individual products, so a product-level score can only indicate relative nutritional quality and cannot judge a person's overall diet. This distinction is carried into the design and the user-facing wording of NutriScan.

2.3 Existing Applications and Systems

Several systems address parts of the problem. Table 2.1 summarizes representative examples. FoodSwitch demonstrated that a barcode-scanning smartphone application backed by a crowdsourced food composition database can offer nutritional information and suggest healthier alternatives [5]. Commercial applications such as Yuka provide scores and alternatives [11], and the Open Food Facts application exposes the open database together with indicators such as Nutri-Score [6].

Table 2.1: Comparison of Representative Food Rating Systems

System	Type	Basis of Rating	Suggests Alternatives
Nutri-Score [1]	Front-of-pack label	Points from energy, sugars, saturated fat, sodium versus fiber, protein and fruit/vegetable content per 100 g; mapped to A to E	No (label only)
Health Star Rating [3]	Front-of-pack label	Algorithm-based rating from half a star to five stars	No (label only)
FoodSwitch [5]	Mobile application	Nutritional information from a crowdsourced composition database	Yes
Yuka [11]	Commercial mobile application	Score out of 100 combining nutritional quality with other factors, as described by the vendor	Yes
Open Food Facts app [6]	Open mobile application	Displays indicators such as Nutri-Score computed from the open database	Via category browsing
NutriScan (proposed)	Mobile application and backend	Transparent rule-based 0.5 to 5 star rating based on the proposed INR, with a breakdown of every score	Yes, ranked within the same category with reasons

2.4 Open Food Facts

2.4.1 Overview

Open Food Facts is a free, collaborative database of food products from around the world, built by contributors who scan products and enter label information [6]. Records are identified by barcode and include product name, brand, categories, ingredients, nutritional values, and derived indicators. The data can be downloaded in bulk, which suits NutriScan's database-driven approach.

2.4.2 Data Quality and Licensing

Because the data are crowdsourced, completeness and accuracy vary: some products lack nutrients, units may be inconsistent, and values may be mistyped. The database is made available under the Open Database License (ODbL), with individual contents under a database contents license, which requires attribution and share-alike conditions for derived databases [7]. NutriScan must therefore clean and validate the data, acknowledge the source, and respect the license terms.

2.5 Comparative Analysis and Research Gaps

The review shows that (i) established nutrient-profiling models agree on a small core of nutrients, (ii) a simple indicator is more useful to consumers than raw tables,

and (iii) barcode-based applications exist but differ in transparency and in how directly they support choosing between products. The following gaps motivate NutriScan:

- Many scores are presented without a step-by-step explanation that a student, shopper or reviewer can trace and verify.
- Comparison across categories using identical thresholds can mislead; category-aware scoring is needed.
- Incomplete data are sometimes scored anyway; an explicit reliability policy that refuses to fabricate values is needed.
- Alternative recommendation is often secondary; making it a primary workflow, with reasons, converts analysis into an actionable shopping decision.

2.6 Motivation and Proposed Direction

NutriScan combines the strengths of prior work: the per-100 g, nutrient-balance logic of established profiling models, the convenience of barcode scanning demonstrated by earlier applications, and the open data of Open Food Facts. It adds a transparent rule-based engine, explicit data-reliability handling, and a ranked, explained alternatives feature at the center of the user experience.

Chapter 3

DATA COLLECTION AND DATASET PREPARATION

This chapter describes the data source used by NutriScan, the fields selected, and the cleaning and preprocessing pipeline that converts raw Open Food Facts records into reliable entries in the PostgreSQL database.

3.1 Data Source

The initial product information is obtained from the Open Food Facts bulk export [6], [7]. The dataset was chosen because it is open, large, identified by barcode, and contains the nutritional fields required by the scoring engine. Every product that NutriScan supports must have a corresponding row in the local database; the application never queries third-party sources at scan time and never invents data for a missing product.

3.2 Selected Fields

The raw export contains a very large number of columns. Only the fields relevant to the application are retained, as listed in Table 3.1.

Table 3.1: Fields Selected from the Open Food Facts Dataset

Source Column	Database Field	Purpose
code	products.barcode	Unique product identifier used for lookup
product_name	products.name	Display and name search
brands	products.brand	Display and filtering
categories_tags	products.category_id	Mapping to application categories
ingredients_text	products.ingredients_text	Ingredient analysis
energy-kcal_100g	nutrition_facts.energy_kcal	Energy score
fat_100g, saturated-fat_100g	nutrition_facts.fat_g, saturated_fat_g	Saturated fat score; total fat display
carbohydrates_100g, sugars_100g	nutrition_facts.carbohydrates_g, sugars_g	Sugar score; display
fiber_100g	nutrition_facts.fiber_g	Fiber score
proteins_100g	nutrition_facts.proteins_g	Protein score
salt_100g, sodium_100g	nutrition_facts.sodium_mg	Sodium score
last_modified_t	products.last_modified	Freshness tracking and updates

3.3 Data Acquisition

The dataset is downloaded as a CSV export and processed offline. Because the file is large, it is processed in chunks. The initial import may be restricted to products sold in the target market (for example, India) and to the product categories supported

in the first release: biscuits and cookies, snacks and chips, chocolates, breakfast cereals, beverages, instant noodles and protein bars. The exact dataset version and download date should be recorded for reproducibility.

3.4 Data Cleaning and Preprocessing

The cleaning pipeline applies the steps in Table 3.2 in order. Each rule is deterministic so that the same input always produces the same cleaned database.

Table 3.2: Data Cleaning and Preprocessing Rules

Step	Operation	Rule
1	Barcode normalization	Keep numeric codes of valid length (EAN-8, UPC-A, EAN-13); convert UPC-A to 13 digits with a leading zero; verify the check digit [16].
2	Duplicate removal	Keep one record per barcode; where duplicates exist, retain the record with the most complete nutritional fields and latest modification time.
3	Unit standardization	Express all nutrients per 100 g (per 100 ml for beverages). Convert kJ to kcal (1 kcal = 4.184 kJ). Convert salt to sodium (sodium = salt / 2.5) and report sodium in mg.
4	Plausibility checks	Reject negative values; reject values above 100 g per 100 g; reject energy above 900 kcal per 100 g; require sugars not greater than carbohydrates and saturated fat not greater than total fat.
5	Missing value handling	Do not impute nutritional values. Mark a field as missing (NULL) and record the completeness of each product.
6	Text cleaning	Trim whitespace, normalize case and character encoding of names and brands; remove records without a usable name.
7	Category mapping	Map hierarchical Open Food Facts category tags to the application's primary categories; unmapped products are excluded from alternative recommendation.
8	Import	Load cleaned records into PostgreSQL inside transactions, using upserts keyed by barcode.

3.5 Category Normalization

Open Food Facts assigns multiple, hierarchical category tags to a product. NutriScan defines a small set of primary categories and maps tags to them using a configurable lookup table. A product receives exactly one primary category, which is used for category-aware scoring thresholds and for retrieving alternatives. Maintaining this mapping as data, rather than code, allows categories to be refined without changing the application logic.

3.6 Data Quality Assessment

After cleaning, the completeness of each nutritional field is measured to decide which products are eligible for scoring and for recommendation. Table 3.3 records the summary statistics of the prepared dataset.

Table 3.3: Summary of the Prepared Dataset

Measure	Value
Raw records in the downloaded export	[N]
Records after barcode validation and de-duplication	[N]
Records after plausibility checks	[N]
Records mapped to a primary category	[N]
Records with all core nutrients (eligible for full score)	[N (P%)]
Records with fiber or protein missing (partial score)	[N (P%)]
Records with a core nutrient missing (no score)	[N (P%)]

3.7 Licensing and Data Currency

Open Food Facts data are subject to the ODbL and associated licenses [7]. NutriScan will attribute the source in the application and in this report, and any redistribution of derived data will follow the share-alike terms. Product formulations change over time, so each record stores its last-modified time and the database is refreshed periodically from newer exports. The application states the data source and encourages users to verify against the physical label.

Chapter 4

ANALYSIS AND DESIGN

This chapter analyzes the requirements of NutriScan and presents its design: the system architecture, the database, the initial Health Score design, the current INR-based score calculation, the recommendation and comparison algorithms, and the API.

4.1 Problem Analysis

A user standing in a shop wants an answer within seconds. The system must therefore be fast, must produce an understandable result, and must explain it. Three questions drive the design: what is in this product, how does it compare nutritionally, and is there a better alternative? Each question maps to a module: product analysis, nutritional rating, and recommendation and comparison.

4.2 Requirement Analysis

4.2.1 Functional Requirements

Table 4.1: Functional Requirements

ID	Requirement
FR1	The system shall scan a product barcode using the phone camera and validate its format.
FR2	The system shall retrieve product information from the database using the barcode.
FR3	The system shall inform the user when a product is not in the database and shall not display invented nutritional values.
FR4	The system shall compute the INR-based final score and star rating (0.5 to 5 stars) using the rule-based engine.
FR5	The system shall display the nutritional breakdown, ingredients, positive attributes, concerns and an explanation of the score.
FR6	The system shall recommend higher-rated alternatives from the same category with reasons.
FR7	The system shall compare two or more products side by side and highlight differences.
FR8	The system shall support search by product name and browsing by category.
FR9	The system shall allow users to register, log in, view scan history, save favorites and set preferences.
FR10	The system shall indicate when data are incomplete and shall not calculate an unreliable score.
FR11	The system shall allow users to submit details of missing products for review.

4.2.2 Non-Functional Requirements

Table 4.2: Non-Functional Requirements

Attribute	Requirement
Performance	Barcode lookup and score display should feel immediate on a typical mobile network; database queries use indexes on barcode, category and score.

Attribute	Requirement
Transparency	Every score must be reproducible from stored values and documented rules.
Reliability	Missing or implausible data must be flagged rather than guessed.
Usability	Results must be understandable without nutrition expertise; color and wording must be clear.
Security and privacy	Passwords hashed, transport over HTTPS, parameterized queries, minimal personal data.
Maintainability	Scoring thresholds and category mappings stored as data and versioned.
Portability	Cross-platform mobile client (Android first, iOS potential).

4.3 Feasibility Analysis

4.3.1 Technical Feasibility

All components use mature, well-documented open-source technologies: React Native with Expo for the client [13], Node.js with Express.js for the backend [14], and PostgreSQL for storage [12]. The Expo ecosystem provides camera and barcode-scanning support, and PostgreSQL supports the indexed category and ranking queries the recommendation feature requires.

4.3.2 Operational Feasibility

The workflow (scan, view result, view alternatives) is simple and matches a common shopping habit. No special hardware is needed beyond a smartphone camera.

4.3.3 Economic Feasibility

The data source and the major frameworks are free and open source. The main costs are modest cloud hosting for the backend and database, which can be kept low for a student project and scaled later.

4.4 System Architecture

NutriScan follows a three-tier architecture, shown in Figure 4.1. The React Native client communicates with the Express backend through a REST interface [15]. The backend contains the product service, scoring engine, recommendation engine, comparison service and authentication and history service. PostgreSQL stores products, nutrition facts, scores, categories and user data. A separate offline ETL pipeline prepares the Open Food Facts data and loads it into the database.

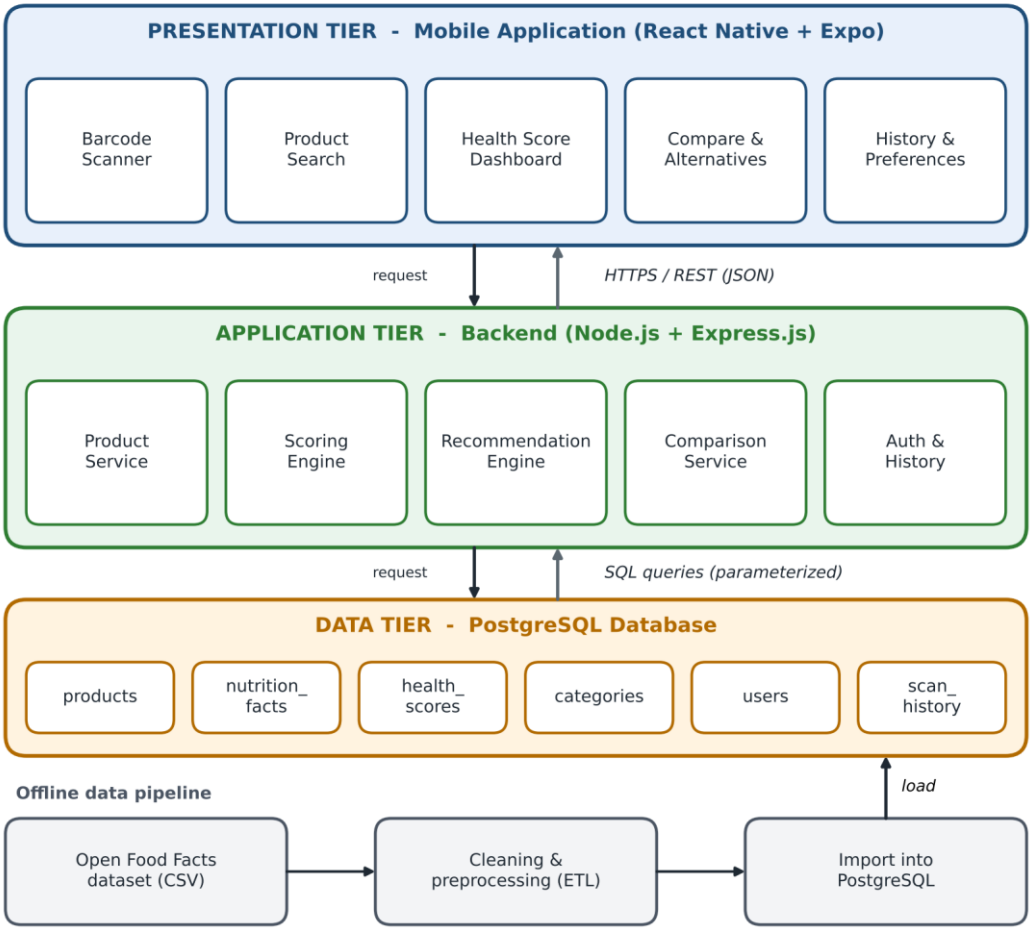


Figure 4.1: Three-Tier System Architecture of NutriScan

4.5 Database Design

The database is normalized so that product identity, nutritional values and computed scores are stored separately. Figure 4.2 shows the entity-relationship diagram and Table 4.3 summarizes the tables. The complete schema is given in Appendix A.

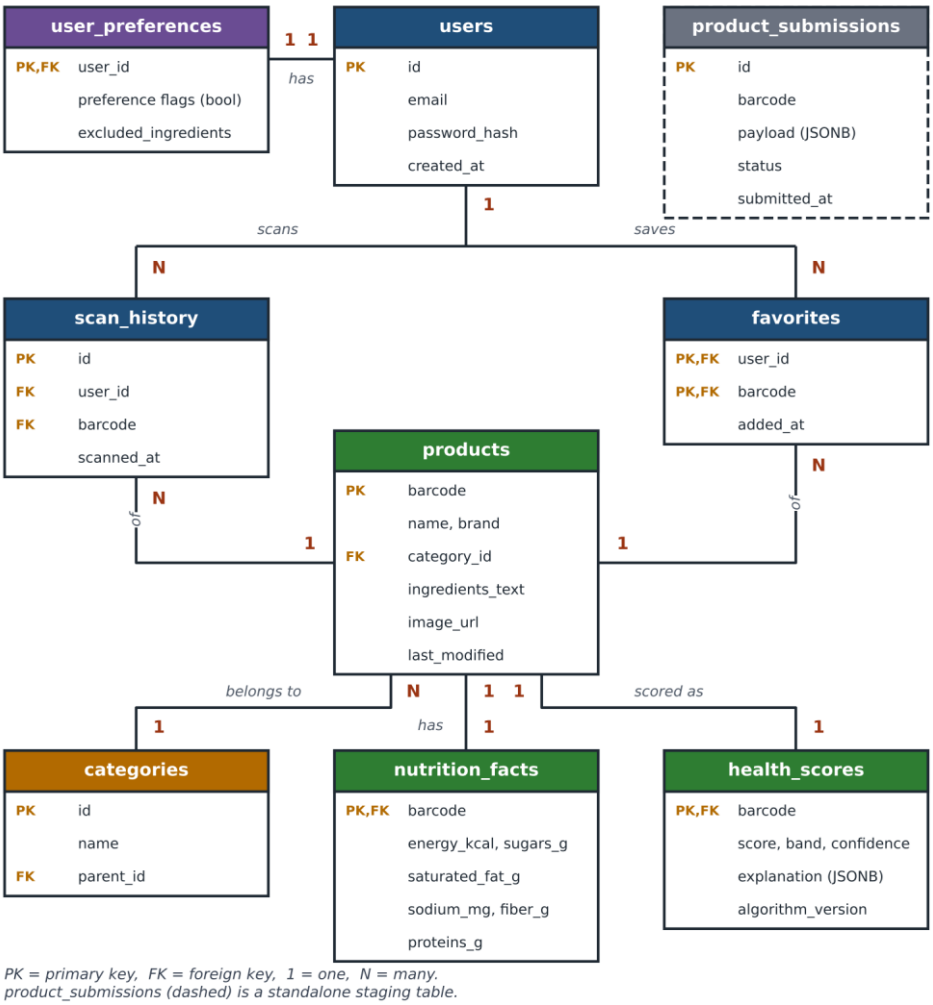


Figure 4.2: Entity-Relationship Diagram of the NutriScan Database

Table 4.3: Summary of Database Tables

Table	Description
products	One row per product: barcode (primary key), name, brand, category, ingredients, image, last-modified time.
nutrition_facts	Nutritional values per 100 g or 100 ml, one row per product; unknown values stored as NULL.
health_scores	Computed score, band, confidence level, explanation (JSON) and algorithm version.
categories	Primary categories and optional parent categories.
category_thresholds	Best and worst thresholds per scoring parameter and category, with a default set.
users, user_preferences	Accounts and preference flags such as lower sugar or excluded ingredients.
scan_history, favorites	Per-user scanned products and saved favorites.
product_submissions	User-submitted missing products awaiting review.

4.6 Initial Health Score Methodology

This section records the 0 to 100 Health Score designed in the initial phase of the project. It has been superseded by the INR-based star rating described in Section 4.7, which is the method NutriScan currently uses, and is retained here to document the design process.

4.6.1 Design Principles

The scoring system follows five principles: it is rule-based and explainable; it uses nutrients with established health guidance [8], [9], [17]; it works on a per-100 g (or 100 ml) basis so products are comparable regardless of pack size [1]; it is category-aware; and it never scores data that are too incomplete to be reliable. The thresholds below are preliminary values chosen for the initial version and must be validated against an independent reference before release.

4.6.2 Score Formulation

For each parameter i with measured value x_i, a normalized position t_i is computed between a best threshold lo_i and a worst threshold hi_i: $t_i = \min(1, \max(0, (x_i - lo_i) / (hi_i - lo_i)))$. For nutrients to limit (sugars, saturated fat, sodium, energy) the sub-score is $s_i = 1 - t_i$. For nutrients to encourage (fiber, protein) the sub-score is $s_i = t_i$, with $lo_i = 0$ and hi_i the amount at which the full benefit is awarded. The Health Score is the weighted sum $S = 100 \times (w_1s_1 + w_2s_2 + ... + w_6s_6)$, rounded to the nearest integer, where the weights sum to 1.

4.6.3 Parameters, Weights and Thresholds

Table 4.4 lists the initial default parameters. Sugars, saturated fat and sodium receive the largest weights because they are the nutrients most consistently associated with guidance to limit intake [8], [9], [17]. Energy density, fiber and protein refine the score.

Table 4.4: Initial Default Scoring Parameters (per 100 g or 100 ml)

Parameter	Direction	Weight	Best (s = 1)	Worst (s = 0)
Sugars (g)	Lower is better	0.25	5 or less	40 or more
Saturated fat (g)	Lower is better	0.20	1 or less	10 or more
Sodium (mg)	Lower is better	0.20	90 or less	900 or more
Energy (kcal)	Lower is better	0.15	100 or less	550 or more
Dietary fiber (g)	Higher is better	0.10	6 or more	0
Protein (g)	Higher is better	0.10	12 or more	0

4.6.4 Category-Aware Thresholds

Comparing biscuits with breakfast cereals or beverages using identical thresholds can mislead. The thresholds are therefore stored per category in the category_thresholds table, with the defaults of Table 4.4 used as a fallback. Category thresholds can be calibrated from the distribution of the cleaned data (for example, using the 10th and 90th percentiles of each nutrient within the category) and reviewed against nutritional guidance. Beverages are evaluated per 100 ml with their own thresholds.

4.6.5 Score Bands

Table 4.5: Preliminary Health Score Bands

Score Range	Interpretation
80 to 100	Better nutritional profile
60 to 79	Relatively good nutritional profile
40 to 59	Moderate nutritional profile
20 to 39	Less favorable nutritional profile
0 to 19	Poor nutritional profile

These ranges are preliminary and are validated against an independent reference before release. The wording avoids medical claims: a band describes a product's nutritional profile relative to the scoring rules, not its suitability for a particular person.

4.6.6 Handling Missing Data

A score is calculated only when energy, sugars, saturated fat and sodium (or salt) are all available. If fiber or protein is missing, the score is calculated over the available parameters with the weights renormalized, and it is labeled as partial; partial scores are shown with a notice and are excluded from alternative recommendations by default. If any core nutrient is missing, the application states that a score cannot be calculated reliably. Missing values are never imputed.

4.6.7 Explanation Generation

For each parameter the engine records the value, sub-score and weighted contribution. Parameters with sub-score of 0.7 or higher are listed as positive attributes, and those with sub-score below 0.3 are listed as concerns (for example, high sugar or high sodium). The explanation is assembled from these records using fixed sentence templates so that it always agrees with the score.

4.7 Current Score Calculation: INR-Based Star Rating

This section describes the method currently used by NutriScan to calculate the nutritional score of a product and convert it into a star rating. It replaces the initial Health Score of Section 4.6.

4.7.1 Overview

NutriScan is a nutrition analysis application designed to help users understand the nutritional composition of packaged food products. It analyses the nutritional information available in the product database and generates a star-based nutritional rating.

The scoring methodology used in NutriScan is based on the principles of the Indian Nutrition Rating (INR) system proposed in the Food Safety and Standards Authority of India (FSSAI) draft notification issued in 2022 [18]. The proposed system expresses the nutritional profile of eligible packaged foods on a scale ranging from 0.5 to 5 stars.

Regulatory note: The implementation in NutriScan is an academic prototype based on the proposed INR methodology. It is not an FSSAI-certified application, and the use of the draft methodology does not establish official approval or regulatory compliance.

4.7.2 Objectives of the Scoring System

The main objectives of the nutrition scoring system are:

1. To evaluate the nutritional composition of packaged food products.
2. To calculate a numerical score using predefined nutrient thresholds.
3. To consider both nutrients of concern and beneficial nutritional components.
4. To convert the final score into an easy-to-understand star rating.
5. To provide users with a transparent explanation of the factors influencing the rating.

4.7.3 Nutritional Data Standardization

For meaningful comparison, nutritional values are considered per 100 grams for solid foods and per 100 millilitres for liquid foods, consistent with the relevant basis described in the proposed methodology.

The product database may contain the following information:

- Energy (kcal)
- Total sugars (g)
- Saturated fat (g)
- Sodium (mg)
- Protein (g)
- Dietary fibre (g)
- Fruit and vegetable percentage
- Nuts, legumes and millets percentage

The reliability of the rating depends on the accuracy and completeness of these nutritional values.

4.7.4 Baseline Point Calculation

The proposed INR methodology assigns baseline points to four nutritional factors:

1. Energy
2. Saturated fat
3. Total sugars
4. Sodium

Each nutrient is evaluated using the applicable threshold table. The nutrient value determines the corresponding points, with higher values generally resulting in more baseline points.

The baseline score is calculated as:

$$Baseline\ Score = Energy\ Points + Saturated\ Fat\ Points + Sugar\ Points + Sodium\ Points$$

Mathematically,

$$B = E + S + U + Na$$

Where:

- B = Baseline score
- E = Energy points
- S = Saturated-fat points

- U = Total-sugar points
- Na = Sodium points

The thresholds are taken from the reference methodology rather than being selected according to personal preference.

4.7.5 Positive Point Calculation

The proposed methodology also considers beneficial nutritional components. These can reduce the baseline score when the product meets the relevant eligibility and threshold requirements. The positive factors are listed in Table 4.6.

Table 4.6: Positive Factors in the INR Score

Factor	Description
FV	Qualifying fruit and vegetable content
NLM	Qualifying nuts, legumes and millets content
Fibre	Dietary fibre content
Protein	Protein content

Each factor is assigned points according to the applicable threshold table. The methodology also specifies limits on positive points, which depend on the baseline score. These limits prevent the positive factors from reducing the score without restriction.

The total positive score is:

$$Positive\ Score = FV\ Points + NLM\ Points + Fibre\ Points + Protein\ Points$$

4.7.6 Final Score Calculation

The final nutritional score is obtained by subtracting the eligible positive points from the baseline score.

$$Final\ Score = Baseline\ Score - Positive\ Score$$

Mathematically,

$$Final\ Score = B - (FV + NLM + F + P)$$

Where:

- B = Baseline score
- FV = Fruit and vegetable points

- *NLM* = Nuts, legumes and millets points
- *F* = Fibre points
- *P* = Protein points

A lower final score corresponds to a higher star rating under the proposed solid-food rating scale.

4.7.7 Conversion of Score into Star Rating

For solid foods, the proposed INR methodology maps the final score to the star ratings given in Table 4.7.

Table 4.7: Proposed INR Star Rating Scale for Solid Foods

Final Score	Proposed Rating
−11 or lower	5 stars
−10 to −7	4.5 stars
−6 to −2	4 stars
−1 to 2	3.5 stars
3 to 6	3 stars
7 to 11	2.5 stars
12 to 15	2 stars
16 to 20	1.5 stars
21 to 24	1 star
25 or higher	0.5 stars

Liquid foods use a separate rating scale. Therefore, the application must identify the product category before converting a score into stars. Product categories that are excluded under the relevant methodology should not automatically be assigned a rating.

4.7.8 Illustrative Example

Consider a hypothetical solid food product with the baseline points shown in Table 4.8.

Table 4.8: Illustrative Baseline Points

Nutritional Factor	Illustrative Points
Energy	3
Saturated fat	3
Total sugars	1
Sodium	4
Total baseline score	**11**

The baseline score is:

$$B = 3 + 3 + 1 + 4 = 11$$

Assume that the product qualifies for the positive points shown in Table 4.9 after applying the relevant eligibility conditions and point caps.

Table 4.9: Illustrative Positive Points

Positive Factor	Illustrative Points
Fruit and vegetables	0
Nuts, legumes and millets	0
Fibre	1
Protein	6
Total positive score	**7**

The final score is:

$$Final\ Score = 11 - 7 = 4$$

According to the proposed solid-food rating scale (Table 4.7), a final score of 4 corresponds to a rating of 3 stars.

This example is illustrative only. Actual product scores must be calculated using the exact applicable nutrient thresholds, positive-point rules and caps from the reference document.

4.7.9 Algorithmic Implementation in NutriScan

The scoring procedure can be implemented as the following sequence:

1. Retrieve the selected product's nutritional information from the database.
2. Identify whether the product is a solid food or a liquid.
3. Validate the nutritional data and standardize it to the appropriate reference quantity.
4. Calculate energy, saturated-fat, sugar and sodium points using the applicable threshold tables.
5. Add these points to obtain the baseline score.
6. Calculate eligible positive points for FV, NLM, fibre and protein.

7. Apply the positive-point caps specified by the reference methodology.
8. Subtract the positive score from the baseline score.
9. Convert the final score into the appropriate star rating.
10. Display the rating, nutrient information and score breakdown to the user.

If required nutritional values are missing or unreliable, the application should indicate that the rating is incomplete or estimated rather than treating unavailable information as zero.

4.7.10 Advantages of the Methodology

- **Structured evaluation:** Uses defined nutrient thresholds rather than subjective judgement.
- **Easy interpretation:** Presents the result through a simple star-rating scale.
- **Transparency:** Allows users to understand how different nutritional factors influence the result.
- **Comparability:** Uses a standardized nutritional reference quantity.
- **Extensibility:** The scoring logic can be updated if the reference methodology changes.

4.7.11 Limitations

The accuracy of the result depends on the completeness and reliability of the nutritional database. Ingredient percentages required for FV and NLM calculations may not always be available. Missing or estimated values can affect the final score.

Furthermore, the implementation is a prototype based on an FSSAI draft methodology. It should not be represented as an official FSSAI rating, certification or determination of whether a product is healthy for every individual. The star rating is a simplified nutritional comparison tool and does not replace professional dietary advice. The draft notification [18] should be consulted for the complete proposed INR tables, eligibility requirements, nutrient thresholds, point caps and rating rules.

4.8 Recommendation Algorithm

The alternatives feature is the most important part of NutriScan. Figure 4.3 shows the end-to-end flow from scan to recommendation. After the scanned product's rating is obtained, the recommendation engine performs the following steps:

1. Identify the primary category of the scanned product.
2. Retrieve other products in the same category with stored star ratings.
3. Filter out products with insufficient nutritional information (incomplete or missing ratings).
4. Remove products that do not satisfy the user's preferences or exclusions, if any are set.
5. Rank the remaining products by star rating, keeping only those rated higher than the scanned product.
6. Return the top results with, for each, the parameters in which it is better (for example, lower sugar or higher fiber).

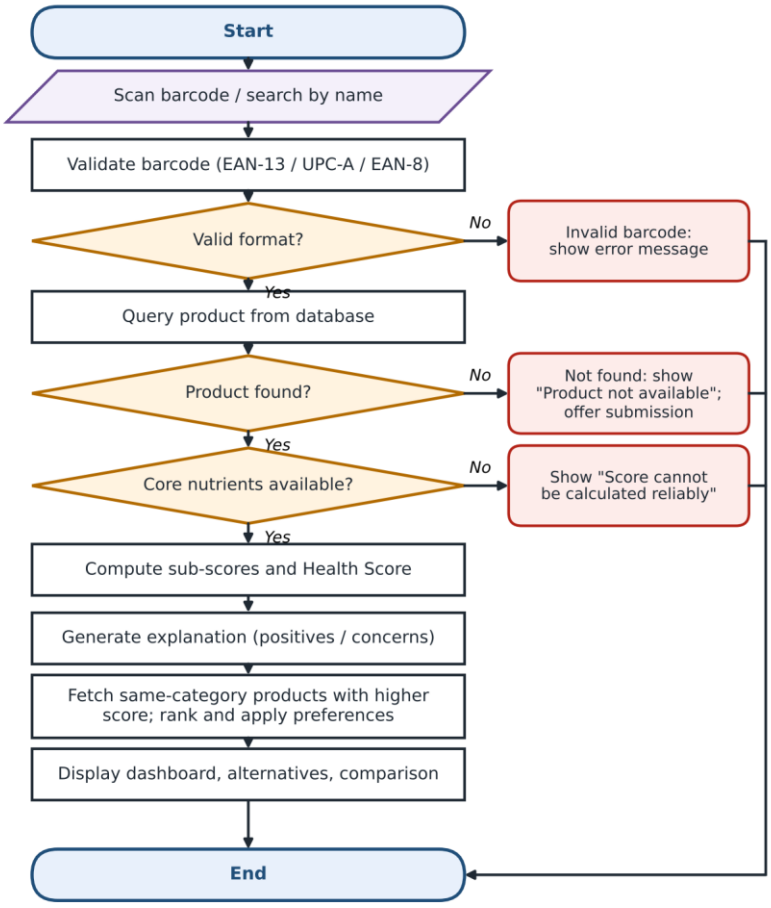


Figure 4.3: Flow from Barcode Scan to Alternative Recommendation

Because ratings are precomputed and stored with an index, retrieving alternatives is a single indexed query. Ranking can be refined by using the rating difference as the primary key and the preference match as a tie-breaker.

4.9 Product Comparison Module

Given two or more barcodes, the comparison service retrieves their nutrition facts and scores, aligns them in a common structure and computes the difference per parameter. The client displays the values side by side, highlights the more favorable value in each row, and shows a one-line summary stating which product has the more favorable profile and why.

4.10 API Design

The backend exposes a RESTful JSON API [15]. Table 4.10 lists the principal endpoints.

Table 4.10: Principal REST API Endpoints

Method	Endpoint	Purpose
GET	/api/products/:barcode	Product details, nutrition, star rating, score breakdown and explanation
GET	/api/products/search?q=	Search products by name
GET	/api/products/:barcode/alternatives	Ranked healthier alternatives in the same category
POST	/api/compare	Compare a list of barcodes
GET	/api/categories	List primary categories
GET	/api/categories/:id/products	Products in a category sorted by score
POST	/api/auth/register, /login	Account creation and authentication
GET, POST	/api/history	Retrieve or add scan history
GET, POST, DELETE	/api/favorites	Manage favorites
GET, PUT	/api/preferences	Read or update preferences
POST	/api/submissions	Submit a missing product for review

4.11 Security and Privacy Considerations

Passwords are stored only as salted hashes, sessions use signed tokens, all traffic uses HTTPS, and database access uses parameterized queries to prevent injection. Rate limiting protects the API from abuse. Scan history and preferences are personal data, so only what is necessary is stored, it is accessible only to its owner, and users can delete it. A guest mode allows scanning without an account.

APPENDIX A: DATABASE SCHEMA

The following SQL creates the principal tables of the NutriScan database.

```
CREATE EXTENSION IF NOT EXISTS pg_trgm;

CREATE TABLE categories (
  id SERIAL PRIMARY KEY,
  name TEXT UNIQUE NOT NULL,
  parent id INT REFERENCES categories(id)
);

CREATE TABLE products (
  barcode VARCHAR(14) PRIMARY KEY,
  name TEXT NOT NULL,
  brand TEXT,
  category id INT REFERENCES categories(id),
  ingredients_text TEXT,
  image_url TEXT,
  source TEXT DEFAULT 'openfoodfacts',
  last_modified TIMESTAMP
);

CREATE TABLE nutrition facts (
  barcode VARCHAR(14) PRIMARY KEY REFERENCES products(barcode),
  basis TEXT CHECK (basis IN ('per_100g','per_100ml')),
  energy_kcal NUMERIC(6,1),
  fat g NUMERIC(5,1),
  saturated fat g NUMERIC(5,1),
  carbohydrates g NUMERIC(5,1),
  sugars_g NUMERIC(5,1),
  fiber_g NUMERIC(5,1),
  proteins_g NUMERIC(5,1),
  sodium mg NUMERIC(7,1)
);

CREATE TABLE health scores (
  barcode VARCHAR(14) PRIMARY KEY REFERENCES products(barcode),
  score SMALLINT CHECK (score BETWEEN 0 AND 100),
  band TEXT,
  confidence TEXT CHECK (confidence IN ('full','partial')),
  explanation JSONB,
  algorithm version TEXT,
  computed at TIMESTAMP DEFAULT now()
);

CREATE TABLE category_thresholds (
  category_id INT REFERENCES categories(id),
  parameter TEXT NOT NULL,
  best NUMERIC NOT NULL,
  worst NUMERIC NOT NULL,
  weight NUMERIC NOT NULL,
  PRIMARY KEY (category_id, parameter)
);

CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen random uuid(),
  email TEXT UNIQUE NOT NULL,
  password hash TEXT NOT NULL,
  created_at TIMESTAMP DEFAULT now()
);

CREATE TABLE user_preferences (
  user_id UUID PRIMARY KEY REFERENCES users(id),
  low sugar BOOLEAN DEFAULT false,
  high protein BOOLEAN DEFAULT false,
  high fiber BOOLEAN DEFAULT false,
  low_sodium BOOLEAN DEFAULT false,
```

```
  vegetarian BOOLEAN DEFAULT false,
  excluded ingredients TEXT[]
);

CREATE TABLE scan history (
  id BIGSERIAL PRIMARY KEY,
  user_id UUID REFERENCES users(id),
  barcode VARCHAR(14) REFERENCES products(barcode),
  scanned at TIMESTAMP DEFAULT now()
);

CREATE TABLE favorites (
  user id UUID REFERENCES users(id),
  barcode VARCHAR(14) REFERENCES products(barcode),
  added_at TIMESTAMP DEFAULT now(),
  PRIMARY KEY (user_id, barcode)
);

CREATE TABLE product submissions (
  id SERIAL PRIMARY KEY,
  barcode VARCHAR(14),
  payload JSONB,
  status TEXT DEFAULT 'pending',
  submitted_at TIMESTAMP DEFAULT now()
);

CREATE INDEX idx_products_category ON products(category_id);
CREATE INDEX idx scores score ON health scores(score DESC);
CREATE INDEX idx_products_name_trgm ON products USING gin (name gin_trgm_ops);
```

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